Resolving CheXpert Uncertainty Labels with Vision-Language Models: Ground-Truth Validation and Impact on Downstream Classification

CSE 472 — Machine Learning Sessional
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1 Introduction

Chest X-ray (CXR) interpretation is one of the most frequently requested diagnostic procedures in clinical medicine. The CheXpert dataset [1], released by Stanford University, is among the most widely used large-scale CXR benchmarks, containing approximately 224,000 frontal-view radiographs annotated for 14 clinically relevant pathological findings. These annotations were generated using a rule-based natural language processing (NLP) labeler applied to the associated radiology reports. While this automated labeling strategy enables the construction of very large datasets at scale, it comes with a well-known limitation: a substantial fraction of labels—approximately 15–20% across the 14 observations—are assigned a special *uncertain* class (encoded as -1) whenever the NLP labeler encounters ambiguous language in the report (e.g., "possible consolidation," "cannot exclude edema").

The existence of these uncertain labels poses a fundamental challenge for supervised learning. Standard approaches in the literature handle uncertain labels in one of four ways: (1) **U-Ignore**, which simply discards all uncertain samples, wasting up to 20% of the training data; (2) **U-Zeros**, which maps all uncertain labels to the negative class, systematically introducing false negatives; (3) **U-Ones**, which maps all uncertain labels to the positive class, systematically introducing false positives; and (4) treating the uncertain label as a separate third class in a multi-class formulation, which does not reflect the binary nature of clinical diagnosis. Each of these strategies either discards valuable training signal or injects systematic bias into the learned model, both of which degrade downstream classifier performance.

A critical insight motivating our work is that uncertainty in radiology reports arises not because the pathology is truly indeterminate, but because the NLP labeler lacks the ability to reason jointly over the visual evidence in the image and the clinical context described in the report. A trained radiologist confronted with the same ambiguous report language would supplement the textual cues with a detailed visual inspection of the radiograph to arrive at a confident binary diagnosis. This is precisely the capability embodied by recent *Vision-Language Models* (VLMs)—neural architectures trained on paired image-text corpora that develop a joint understanding of visual and linguistic content.

VLMs such as BioViL-T [2], CheXzero [3], BioMedCLIP [4], and WhyXrayCLIP are uniquely suited to the uncertainty resolution task for several reasons. First, they encode rich visual features from the CXR image that allow them to detect subtle radiological patterns associated with each pathology. Second, their text encoders have been pre-trained on large biomedical corpora, endowing them with clinical knowledge that purely visual models lack. Third, the contrastive image-text alignment objective used during pre-training enables a natural zero-shot classification paradigm: given positive and negative clinical prompt descriptions, the model scores the image against each prompt and assigns the label corresponding to the more similar prompt. This approach can be extended to a few-shot setting by constructing prototype embeddings from labeled support examples.

Despite the strong motivation for applying VLMs to uncertainty resolution, two important gaps remain in the existing literature. First, no prior work has systematically benchmarked a diverse suite of chest-X-ray VLMs specifically on uncertain samples from CheXpert, comparing zero-shot and few-shot strategies alongside ablations over different context-enrichment approaches. Second, and more critically, no existing work has validated VLM-generated uncertainty resolutions against a multi-radiologist expert consensus, making it impossible to determine whether VLM relabelings are clinically trustworthy.

This project addresses both gaps through a four-stage pipeline: (1) benchmarking individual VLMs on a radiologist-annotated ground-truth test set; (2) constructing ensemble models and validating their relabelings on the same ground-truth set; (3) relabeling all uncertain samples in the CheXpert training set using the best ensemble; and (4) training downstream chest pathology classifiers on the relabeled data and measuring the impact on classification performance relative to standard uncertainty-handling baselines.

2 Ground-Truth Test Set (GT-Set)

All VLM benchmarking in this project is performed against a high-quality radiologist-annotated test set. When the CheXpert dataset was originally collected, a dedicated test set of 500 studies from 500 distinct patients was assembled, randomly sampled from 1,000 studies in the report test split. This test set was independently annotated by eight board-certified radiologists following the same procedure and post-processing protocol used for the validation set.

The final ground-truth label for each study is determined by the **majority vote of five** radiologist annotations; three of these five radiologists were the same individuals who annotated the validation set, and the remaining two were randomly sampled from the panel of eight. The annotations from the three remaining radiologists were used to benchmark individual radiologist performance.

This majority-vote expert consensus constitutes what we term the **GT-set** throughout this report. Because the GT-set provides reliable binary ground-truth labels (0 = absent, 1 = present) for each study, it serves as the evaluation benchmark for all VLM benchmarking experiments described in the following sections.

3 Pathology Selection

The CheXpert dataset covers 14 clinical observations. Rather than attempting to resolve uncertainty across all 14 labels simultaneously, we focused on five target pathologies: **Edema**, **Atelectasis**, **Pleural Effusion**, **Enlarged Cardiomediastinum**, and **Consolidation**.

The selection was driven by a systematic analysis of the distribution of uncertain labels (-1) in the training set and the corresponding coverage in the GT-set. Table 1 shows the percentage of uncertain samples per label in both the CheXpert training set and the GT-set, sorted by training-set uncertainty percentage.

Table 1: Percentage of uncertain (-1) samples per pathology in the CheXpert training set and GT-set, sorted by training uncertainty. Selected labels are highlighted.

Pathology	**Train Uncertain (%)**	**GT Uncertain (%)**	**Status**
Pneumonia	67.99	71.70	Excluded (too high)
Atelectasis	49.30	53.64	**Selected**
Pleural Other	40.87	45.00	Excluded (no GT coverage
Consolidation	39.28	33.16	**Selected**
Enlarged Cardiomediastinum	27.66	29.82	**Selected**
Cardiomegaly	17.50	9.52	Excluded (low GT coverag
Edema	15.11	14.53	**Selected**
Lung Lesion	12.46	16.00	Excluded (low GT count)
Pleural Effusion	8.73	10.88	**Selected**
Fracture	5.26	3.92	Excluded (too low)
Lung Opacity	4.75	1.72	Excluded (too low)
Pneumothorax	3.98	2.42	Excluded (too low)
Support Devices	0.88	1.70	Excluded (too low)
No Finding	0.00	0.00	Excluded (zero)

Figure 1 visualises the uncertain-sample percentages for both splits. Labels with *very high* uncertainty (e.g., Pneumonia at ∼68%) were excluded because the GT-set contained insufficient positive or negative GT-confirmed examples to reliably evaluate model performance on those classes. Labels with *very low* uncertainty percentage were also excluded because the absolute count of uncertain samples to be relabeled would be negligible and would have minimal impact on downstream classifier training. The five selected pathologies (**Edema, Atelectasis, pleural effusion , enlarged cardiomediastinum, and consolidation**) occupy a **mid-range uncertainty band** in the training distribution, ensuring that (a) there are enough GT-set uncertain examples to obtain statistically meaningful benchmark metrics, and (b) the relabeled samples represent a sufficiently large fraction of the training data to influence downstream classifier performance.

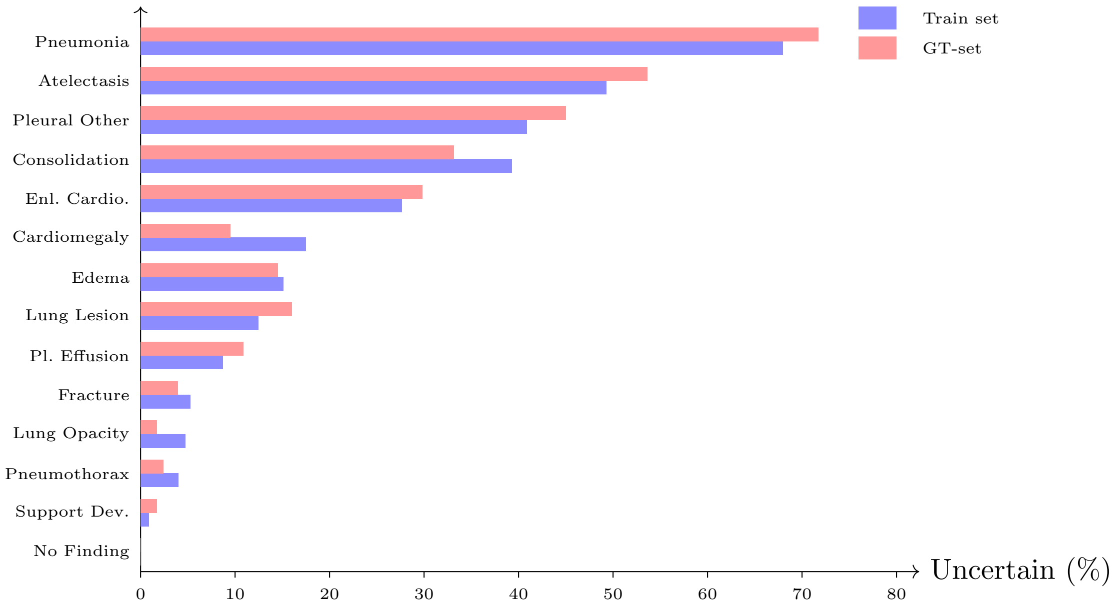

Figure 1: Uncertain-label percentage per pathology for the CheXpert training set (blue) and the GT-set (red)

4 Task Overview

The project is organised into four sequential tasks:

Task 1 Individual VLM benchmarking (zero-shot and few-shot) on the GT-set, with context-format ablation and threshold calibration.

Task 2 Construction of ensemble models (Ensemble A and Ensemble B) and their benchmarking on the GT-set, including a comparison of three ensemble strategies.

Task 3 Relabeling all uncertain samples in the CheXpert training set using the best ensemble configuration.

Task 4 Training downstream multi-label classifiers (DenseNet-121 and EfficientNet-B4) on the relabeled data and comparing against standard baselines.

5 Task 1: Individual VLM Benchmarking

5.1 Models

Four vision-language models were used as individual classifiers:

- **WhyXrayCLIP**: A CLIP-based model trained on 243,334 MIMIC-CXR chest X-rays paired with GPT-4-cleaned radiology reports. It was not trained on CheXpert, ensuring no data leakage. It uses the OpenCLIP API for inference.
- **CheXzero**: A zero-shot chest X-ray model built on top of OpenAI's CLIP architecture, specialised for chest X-ray interpretation via contrastive pre-training on paired CXR-report data.

- **BioMedCLIP**: A CLIP-based model trained on approximately 15 million biomedical image-text pairs sourced from scientific literature (PubMed). Unlike WhyXray-CLIP and CheXzero, BioMedCLIP is not specifically trained on chest X-ray data; it learns general biomedical visual-language associations. Because of this domain gap with chest X-ray diagnosis, we fine-tuned BioMedCLIP on the MIMIC-CXR dataset (see Section 5.7).
- **BioViL-T**: Microsoft's Biomedical Vision-Language Transformer, trained on chest X-ray images and associated radiology reports from MIMIC-CXR. BioViL-T uses a dedicated temporal model that can incorporate prior imaging studies; in our experiments we use it in the single-image configuration with its text encoder.

5.2 Individual Model Benchmarking Pipeline

Figure 2 shows the general benchmarking pipeline applied to every individual model. The pipeline proceeds as follows:

1. **Zero-Shot (ZS) Benchmarking**: The model is evaluated on the GT-set using only positive/negative clinical prompts (with the winning context strategy from the ablation study). A decision threshold calibrated on 1,000 training-set samples is applied.
2. **Few-Shot (FS) Sweep**: An N-shot $\times$ α sweep is performed across all combinations of $n_{\text{shots}} \in \{4, 8, 16, 32, 64\}$ and a grid of α values. For each configuration the combined score threshold is recalibrated. The best configuration per label is selected by maximising $\text{Score} = (\text{AUC} + \text{BAC})/2$ on the calibration split.
3. **Best-Config FS Benchmarking**: The winning configuration from the sweep is applied to the GT-set to produce the final few-shot benchmark numbers.
4. **ZS vs. FS Comparison**: Zero-shot and few-shot results are compared. The better-performing strategy (by mean AUC and BAC across labels) is forwarded to the ensemble stage.

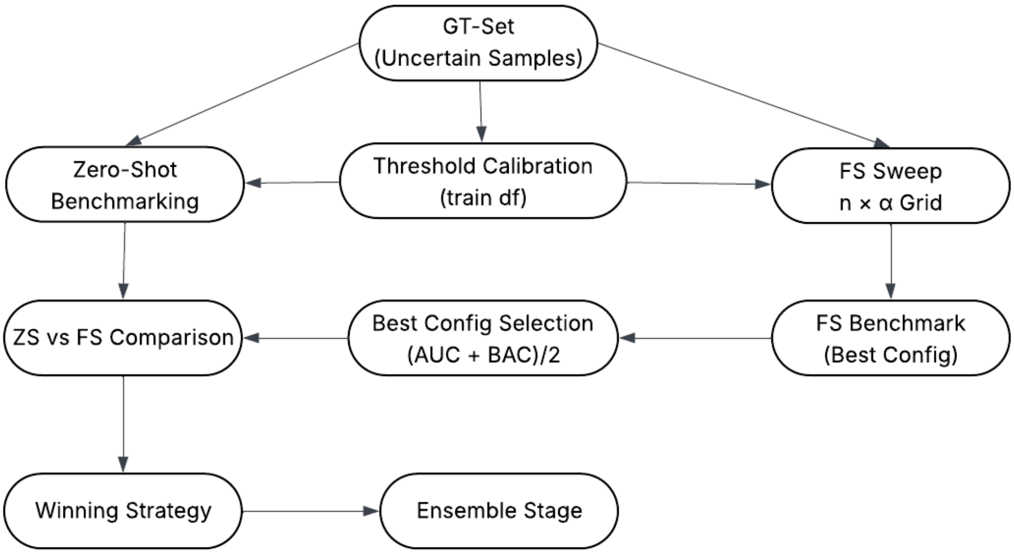

Figure 2: General benchmarking pipeline applied to every individual VLM.

5.3 Feature Correlation Analysis and Context Strategy

Instead of presenting only the image, we perform a **feature correlation analysis** on the CheXpert training set to identify, for each target label, the top-5 most correlated co-occurring pathology features using point-biserial correlation. These are termed **context features**. Three context-format strategies were evaluated in an ablation study for each model:

Plain Text (old_format) Context features are conveyed as simple binary statements: value 1 → "[feature] is present"; value 0 or −1 → "[feature] is absent".

Radiology Format Feature values are translated into natural clinical radiology language (e.g., "bilateral perihilar opacities are present" for Edema), matching the language style of the models' pre-training corpora.

No Context (no_context) Only positive and negative clinical prompts are provided alongside the image, with no co-occurrence context features.

The winning context strategy per label and per model was determined by balanced accuracy on the calibration split, and was then locked in before threshold calibration and benchmarking.

5.4 Prompt Design

For each model, task-specific positive and negative prompts were crafted to match the language style of that model's pre-training corpus. Table 2 summarises the winning context strategy selected for each model–label pair, and Table 3 provides illustrative examples of the positive prompts used per model for two representative labels.

Table 2: Winning context strategy selected per model and per label after the ablation study. *old_format* = plain binary text; *rad_format* = radiology language; *no_context* = no context features.

Model	Edema	Atelectasis	Pl. Effusion	Enl. Cardio.	Consolidation
WhyXrayCLIP	old_format	no_context	no_context	rad_format	old_format
CheXzero	rad_format	rad_format	rad_format	old_format	rad_format
BioViL-T	rad_format	rad_format	old_format	no_context	old_format
BioMedCLIP	old_format	no_context	no_context	no_context	no_context

Table 3: Illustrative examples of positive prompts used for **Edema** and **Atelectasis** across models. Each model has a corresponding negative prompt that describes the absence of the finding.

Model	Positive Prompt – Edema	Positive Prompt – Atelectasis
WhyXrayCLIP	"chest radiograph showing pulmonary edema with symmetric bilateral perihilar bat-wing opacities, vascular congestion and interstitial thickening"	"chest radiograph demonstrating atelectasis with linear or plate-like opacities, volume loss and ipsilateral diaphragmatic elevation"
CheXzero	"chest x-ray showing pulmonary edema with bilateral interstitial opacities perihilar haziness and vascular congestion"	"chest x-ray demonstrating atelectasis with areas of linear opacity volume loss and basilar opacification"
BioViL-T	"pulmonary edema with bilateral perihilar opacities, interstitial thickening, and vascular redistribution"	"atelectasis with linear or subsegmental opacities, volume loss, and elevated hemidiaphragm"
BioMedCLIP	"bilateral pulmonary edema with interstitial opacities, perihilar haziness and pleural effusions"	"pulmonary atelectasis showing linear opacities and basilar volume loss"

5.5 Threshold Calibration

Each VLM produces a continuous similarity score. For CLIP-based models (WhyXray-CLIP, CheXzero, BioMedCLIP) this is $\delta = \mathrm{sim}_{\mathrm{pos}} - \mathrm{sim}_{\mathrm{neg}}$; for BioViL-T it is a normalised cross-attention score. Thresholds were calibrated using 1,000 samples drawn from the CheXpert training set for which the NLP labeler assigned confident binary labels (0 or 1). Three threshold-selection strategies were compared: **Midpoint** ($\tau = 0$), **Median Positive** (median score of positive samples), and **Balanced Accuracy** (exhaustive search maximising balanced accuracy via half-split cross-validation). The Balanced Accuracy strategy was adopted as the selection criterion, as it is robust to class imbalance.

5.6 Zero-Shot Benchmarking Results

Tables 4–7 report the zero-shot benchmark results on the GT-set for all four models.

Table 4: WhyXrayCLIP zero-shot benchmark on GT-set.

Label	AUC	AUPRC	BAC	F1	Sens	Spec
Edema	0.750	0.684	0.750	0.812	1.000	0.500
Atelectasis	0.635	0.621	0.635	0.731	0.864	0.405
Pleural Effusion	0.805	0.600	0.805	0.741	0.909	0.700
Enlarged Cardiomediastinum	0.676	0.641	0.676	0.593	0.471	0.882
Consolidation	0.750	0.091	0.750	0.167	1.000	0.500
Average	**0.723**	**0.527**	**0.723**	**0.609**	**0.849**	**0.598**

Table 5: CheXzero zero-shot benchmark on GT-set.

Label	AUC	AUPRC	BAC	F1	Sens	Spec
Edema	0.712	0.655	0.712	0.774	0.923	0.500
Atelectasis	0.666	0.644	0.666	0.735	0.818	0.514
Pleural Effusion	0.709	0.498	0.709	0.643	0.818	0.600
Enlarged Cardiomediastinum	0.676	0.615	0.676	0.703	0.765	0.588
Consolidation	0.733	0.086	0.733	0.158	1.000	0.467
Average	**0.699**	**0.500**	**0.699**	**0.602**	**0.865**	**0.534**

Table 6: BioMedCLIP (original) zero-shot benchmark on GT-set.

Label	AUC	BAC	F1	Sens	Spec
Edema	0.712	0.712	0.774	0.923	0.500
Atelectasis	0.540	0.540	0.500	0.432	0.649
Pleural Effusion	0.543	0.543	0.483	0.636	0.450
Enlarged Cardiomediastinum	0.618	0.618	0.683	0.824	0.412
Consolidation	0.725	0.725	0.154	1.000	0.450
Average	**0.628**	**0.628**	**0.519**	**0.763**	**0.492**

Table 7: BioViL-T zero-shot benchmark on GT-set.

Label	AUC	AUPRC	BAC	F1	Sens	Spec
Edema	0.715	0.662	0.715	0.759	0.846	0.583
Atelectasis	0.651	0.630	0.651	0.768	0.977	0.324
Pleural Effusion	0.780	0.567	0.780	0.714	0.909	0.650
Enlarged Cardiomediastinum	0.706	0.634	0.706	0.750	0.882	0.529
Consolidation	0.750	0.091	0.750	0.167	1.000	0.500
Average	**0.720**	**0.517**	**0.720**	**0.632**	**0.923**	**0.517**

5.7 BioMedCLIP Fine-Tuning

As noted in Section 5.1, BioMedCLIP was pre-trained on general biomedical literature, not specifically on chest X-ray data, creating a domain gap relative to CheXpert. To close this gap, we fine-tuned BioMedCLIP on MIMIC-CXR using a three-stage curriculum with Low-Rank Adaptation (LoRA) [5].

5.7.1 LoRA Adaptation

LoRA introduces trainable low-rank decomposition matrices $\Delta W = BA$ ($B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, $r \ll \min(d, k)$) into the self-attention linear layers of both the image and text encoders, while keeping the original pre-trained weights frozen. We used rank $r = 16$, scaling $\alpha_{\text{LoRA}} = 32$, and dropout $= 0.05$.

5.7.2 Three-Stage Curriculum

Stage 1 – General CXR Domain Adaptation Trained on MIMIC-CXR frontal images paired with their original radiology reports, using all pathologies. Augmented texts are included to double the effective dataset size.

Stage 2 – Prompt-Aligned Label Supervision Samples are filtered to MIMIC-CXR studies with clearly confirmed positive/negative findings for the five target labels. Each image is paired with the model's own inference-time positive/negative prompt, closing the train–inference distribution gap.

Stage 3 – Uncertainty-Focused Fine-Tuning Samples from MIMIC-CXR whose reports contain ambiguity markers ("possible", "cannot exclude", "likely", "may represent") are used, mirroring the language patterns that trigger the CheXpert NLP labeler to assign -1.

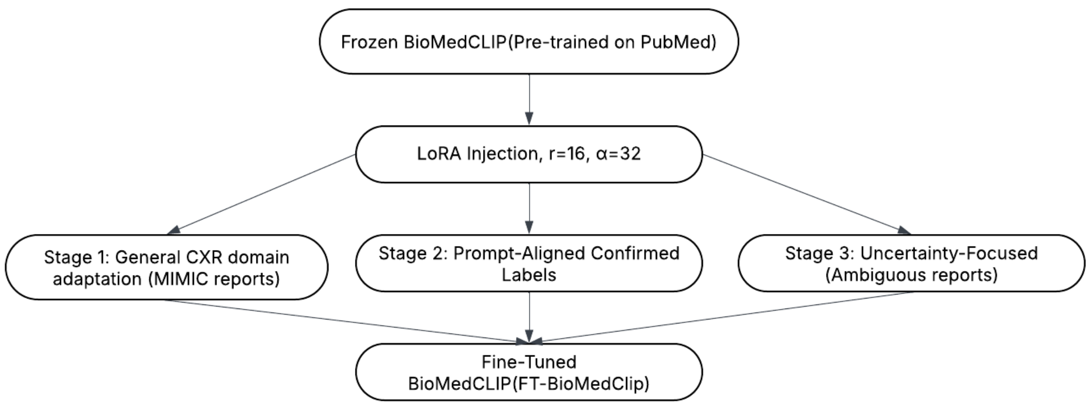

Figure 3: Three-stage curriculum fine-tuning of BioMedCLIP using LoRA. Frozen pre-trained weights are augmented with low-rank trainable matrices across three progressively specialised stages.

5.7.3 Per-Label Model Selection for BioMedCLIP

After fine-tuning, the zero-shot performance of the original and fine-tuned models were compared on a per-label basis. Table 8 shows that fine-tuning improved AUC for Atelectasis (+0.101), Pleural Effusion (+0.034), and Enlarged Cardiomediastinum (+0.147), while it regressed for Edema (-0.122) and Consolidation (-0.108). Therefore, a **per-label model routing** strategy was adopted: for each target label, whichever variant (original or fine-tuned) achieved higher balanced accuracy on the calibration split was selected. This per-label selection approach is unique to BioMedCLIP; for all other individual models, a single best configuration is applied uniformly across labels.

Table 8: BioMedCLIP before vs. after fine-tuning (zero-shot AUC on GT-set). The selected model variant per label is shown in the last column.

Label	**Before (AUC)**	**After (AUC)**	**Δ**	**Selected**
Edema	0.712	0.638	−0.122	Original
Atelectasis	0.540	0.641	+0.101	FT
Pleural Effusion	0.543	0.577	+0.034	FT
Enlarged Cardiomediastinum	0.618	0.765	+0.147	FT
Consolidation	0.725	0.617	−0.108	Original

The zero-shot benchmark after applying the smart per-label routing is reported in Table 9.

Table 9: BioMedCLIP smart zero-shot benchmark (per-label model routing) on GT-set.

Label	**AUC**	**AUPRC**	**BAC**	**F1**	**Sens**	**Spec**
Edema	0.724	0.692	0.724	0.696	0.615	0.833
Atelectasis	0.600	0.601	0.600	0.644	0.659	0.541
Pleural Effusion	0.548	0.379	0.548	0.462	0.545	0.550
Enlarged Cardiomediastinum	0.647	0.593	0.647	0.667	0.706	0.588
Consolidation	0.817	0.120	0.817	0.214	1.000	0.633

5.8 Few-Shot Sweep Strategy

WhyXrayCLIP and CheXzero. For both these models, the few-shot pipeline builds mean-pooled positive and negative prototype embeddings from n_{shots} confirmed training samples per class. The combined score is:

$$s_{\text{combined}} = \alpha \cdot s_{\text{text}} + (1-\alpha) \cdot s_{\text{proto}}, \qquad s_{\text{proto}} = \text{sim}(\mathbf{v}, \mathbf{p}^{+}) - \text{sim}(\mathbf{v}, \mathbf{p}^{-}) \tag{1}$$

A sweep over $n_{\text{shots}} \in \{4, 8, 16, 32, 64\}$ and $\alpha \in \{0.00, 0.25, 0.50, 0.75\}$ was run, selecting the configuration that maximises (AUC+BAC)/2 on the calibration split. For CheXzero the optimal α (denoted $\alpha = 1 - \text{txt\_w}$) is additionally swept per label. The best overall configuration found for **WhyXrayCLIP** was **64-shot,** $\alpha = 0.75$ (uniform across labels). For **CheXzero**, **8-shot** with per-label text weights was selected (Edema/Atelectasis: txt_w = 0.2; Pleural Effusion: 0.5; Enlarged Cardiomediastinum: 0.4; Consolidation: 0.3).

BioViL-T. The sweep compared text-only ($\alpha = 1.0$) against various hybrid configurations (Table 10). The **text_only** strategy ($\alpha = 1.0$, no image prototypes) achieved the highest mean score of 0.767 across all labels, outperforming all hybrid few-shot configurations.

Table 10: Top-5 BioViL-T sweep results, ranked by mean Score = (AUC + BAC)/2. The full sweep tested $n \in \{4, 8, 16, 32, 64\}$ and $\alpha \in \{0.00, 0.25, 0.50, 0.75\}$.

Configuration	Mean Score	Δ vs text_only
text_only ($\alpha = 1.0$)	**0.767**	baseline **(Winner)**
hybrid_64_α0.25	0.760	−0.007
hybrid_8_α0.25	0.759	−0.008
hybrid_64_α0.75	0.758	−0.010
hybrid_32_α0.25	0.750	−0.017

BioMedCLIP. The sweep additionally includes a `model` dimension (original vs. ft1) and a `context_strategy` dimension, selecting the best combination per label. The resulting best per-label configurations are summarised in Table 11.

Table 11: BioMedCLIP best per-label few-shot configuration from the sweep.

Label	n	α	AUC	BAC	Model
Edema	4	0.50	0.955	0.878	Original
Atelectasis	4	0.50	0.711	0.654	FT
Pleural Effusion	4	0.50	0.695	0.643	FT
Enlarged Cardiomediastinum	64	0.00	0.785	0.735	FT
Consolidation	32	0.25	0.972	0.808	Original

5.9 Few-Shot Benchmark Results and ZS vs. FS Comparison

Tables 12 and 13 compare zero-shot and best-config few-shot results for WhyXrayCLIP and CheXzero respectively.

Table 12: WhyXrayCLIP: Zero-Shot vs. Few-Shot (64-shot, α=0.75) on GT-set.

Label	AUC		BAC		F1	
	ZS	FS	ZS	FS	ZS	FS
Edema	0.750	**0.910**	0.750	0.708	0.812	0.788
Atelectasis	0.635	**0.736**	0.635	**0.669**	0.731	**0.766**
Pleural Effusion	0.805	**0.923**	0.805	0.784	0.741	0.720
Enlarged Cardiomediastinum	0.676	**0.862**	0.676	**0.794**	0.593	**0.800**
Consolidation	0.750	**0.972**	0.750	0.733	0.167	0.158
Average	0.723	**0.881**	0.723	**0.738**	0.609	**0.646**
						Verdict: Few-Shot better ✓

Table 13: CheXzero: Zero-Shot vs. Few-Shot (8-shot, per-label α) on GT-set.

Label	**AUC**		**BAC**		**F1**	
	ZS	FS	ZS	FS	ZS	FS
Edema	0.712	**0.827**	0.712	**0.750**	0.774	**0.812**
Atelectasis	0.666	**0.749**	0.666	**0.668**	0.735	0.729
Pleural Effusion	0.709	**0.927**	0.709	**0.850**	0.643	**0.786**
Enlarged Cardiomediastinum	0.676	**0.758**	0.676	**0.706**	0.703	**0.737**
Consolidation	0.733	**0.978**	0.733	0.700	0.158	0.143
Average	0.699	**0.848**	0.699	**0.735**	0.602	**0.641**
						Verdict: Few-Shot better ✓

For **BioViL-T**, the text_only strategy was already the winner of the sweep (Table 10), so it serves as both the zero-shot and the best-strategy result. For **BioMedCLIP**, the per-label few-shot configuration in Table 11 substantially outperforms the zero-shot baseline for most labels, with Edema achieving AUC = 0.955 (vs. 0.712 ZS) using the original model with 4-shot α=0.50.

6 Task 2: Ensemble Models

Having benchmarked the individual models, we constructed ensemble configurations to obtain more reliable uncertainty resolutions by combining complementary model predictions. We explored three broad ensemble strategies, described below.

6.1 Ensemble Strategy Overview

Strategy 1: Train-Set Calibration with GT-Set Consolidation Patch. Thresholds for Edema, Atelectasis, Enlarged Cardiomediastinum, and Pleural Effusion are calibrated using 1,000 balanced samples from the CheXpert training set only (50:50 class-balanced). For Consolidation, the default training-set calibration produces a threshold that is too low, generating an excessive number of false positives due to the highly imbalanced class ratio in the training set ($\sim$34.5% positive). A **GT-set threshold scan** is therefore applied for Consolidation: after scoring the 63 GT-set uncertain samples, the threshold is swept from 0.50 to 0.94 and the highest threshold that retains all three true positives is selected ($\tau^* = 0.82$), reducing false positives from 28 to 6 while maintaining sensitivity at 1.0.

Strategy 2: Weighted GT + Train-Set Calibration (Current Method). Thresholds for all five labels are calibrated using a combined pool of GT-set uncertain samples (weight 3$\times$) and class-balanced training-set samples (weight 1$\times$). The 3:1 weighting prioritises the uncertain-case distribution of the GT-set. This is the strategy used in the final Ensemble A and Ensemble B benchmarks reported below.

Strategy 3: Selective Per-Label Ensemble with GT-Set-Only Calibration. A selective ensemble is constructed by including only the strongest-performing models per label based on Ensemble B benchmarking. Thresholds are calibrated exclusively on GT-set uncertain samples to directly optimize performance on ambiguous cases, with a bias toward high sensitivity for imbalanced labels.

Strategy 4: Selective Per-Label Ensemble with Train-Set-Only Calibration. Uses the same per-label model selection as Strategy 3, but thresholds are calibrated using balanced certain samples from the training set. This evaluates whether calibration on confident samples generalizes to uncertain cases.

6.2 Strategy 1

This strategy uses a three-model ensemble comprising **WhyXrayCLIP** (zero-shot), **CheXzero** (10-shot combined text+image anchor with per-label (txt_w, img_w) from `find_best_weights`), and **BioViL-T** (16-shot hybrid, α=0.25). For context strategy, *old_format* is used for WhyXrayCLIP and CheXzero, and *radiology_format* for BioViL-T. Table 14 summarises the per-label model configurations. The same performance-weighted rank fusion mechanism is used.

Table 14: Strategy 1 ensemble model configuration per label. WhyXrayCLIP: zero-shot; CheXzero: 10-shot combined anchor; BioViL-T: 16-shot hybrid α=0.25.

Label	**Model**	**Shots**	α	**Context**	**Weight**
Edema	WhyXrayCLIP	ZS	1.0	old_format	0.712
	CheXzero	10	0.8	old_format	0.708
	BioViL-T	16	0.25	radiology_format	0.769
Atelectasis	WhyXrayCLIP	ZS	1.0	old_format	0.670
	CheXzero	10	0.8	old_format	0.639
	BioViL-T	16	0.25	radiology_format	0.819
Pleural Effusion	WhyXrayCLIP	ZS	1.0	old_format	0.634
	CheXzero	10	0.5	old_format	0.850
	BioViL-T	16	0.25	radiology_format	0.795
Enl. Cardiomediastinum	WhyXrayCLIP	ZS	1.0	old_format	0.765
	CheXzero	10	0.6	old_format	0.647
	BioViL-T	16	0.25	radiology_format	0.782
Consolidation	WhyXrayCLIP	ZS	1.0	old_format	0.700
	CheXzero	10	0.7	old_format	0.700
	BioViL-T	16	0.25	radiology_format	0.675

Table 15 reports the Strategy 1 ensemble benchmark on the GT-set.

Table 15: Strategy 1 ensemble (WhyXrayCLIP ZS + CheXzero 10-shot + BioViL-T 16-shot, α=0.25) benchmark on GT-set.

Label	AUC	BAC	F1	Sens	Spec	n
Edema	0.821	0.718	0.741	0.769	0.667	25
Atelectasis	0.779	0.709	0.753	0.795	0.622	81
Pleural Effusion	0.845	0.759	0.692	0.818	0.700	31
Enl. Cardiomediastinum	0.841	0.706	0.706	0.706	0.706	34
Consolidation	0.989	0.767	0.176	1.000	0.533	63
Macro Avg.	**0.855**	**0.732**	**0.614**	**0.818**	**0.646**	234

6.3 Strategy 2

Two ensemble configurations were evaluated:

- **Ensemble A**: Three-model ensemble combining WhyXrayCLIP (64-shot, α=0.75), CheXzero (8-shot, per-label α), and BioViL-T (text_only). Each model uses its best configuration determined by individual benchmarking.
- **Ensemble B**: Four-model ensemble extending Ensemble A by adding FT-BioMedCLIP using its per-label best configuration from Table 11. Notably, for Enlarged Cardiomediastinum, FT-BioMedCLIP uses α=0.00 (pure prototype score, no text), reflecting the superior prototype-based performance for that label.

The performance-weighted rank fusion score is:

$$w_{m,\ell} = \frac{\mathrm{AUC}_{m,\ell} + \mathrm{BAC}_{m,\ell}}{2}, \qquad s_{\mathrm{ens}}^{(\ell)} = \sum_m w_{m,\ell} \cdot \mathrm{rank}_m(s_m^{(\ell)}) \tag{2}$$

Table 16 lists the per-model, per-label weights used.

Table 16: Performance weights $w = (\mathrm{AUC} + \mathrm{BAC})/2$ per model and label used for rank fusion. Weights are derived from each model's best-config benchmark.

Label	WhyXrayCLIP	CheXzero	BioViL-T	FT-BioMedCLIP
Edema	0.809	0.789	0.733	0.917
Atelectasis	0.703	0.709	0.712	0.683
Pleural Effusion	0.854	0.889	0.811	0.669
Enlarged Cardiomediastinum	0.828	0.732	0.748	0.760
Consolidation	0.853	0.839	0.834	0.890

6.3.1 Ensemble Threshold Calibration

Before benchmarking on the GT-set, thresholds for the ensemble fused scores are calibrated using Strategy 2 (weighted combined pool). For each label:

1. All GT-set uncertain samples with valid binary GT labels are scored by the ensemble.

2. A class-balanced subset of certain (0 or 1) samples from the training set is scored; size per class is $\max(50, n_{\mathrm{GT}} \times 1)$.
3. GT samples receive weight 3.0; training-set samples receive weight 1.0.
4. The threshold maximising weighted balanced accuracy is found via half-split cross-validation.

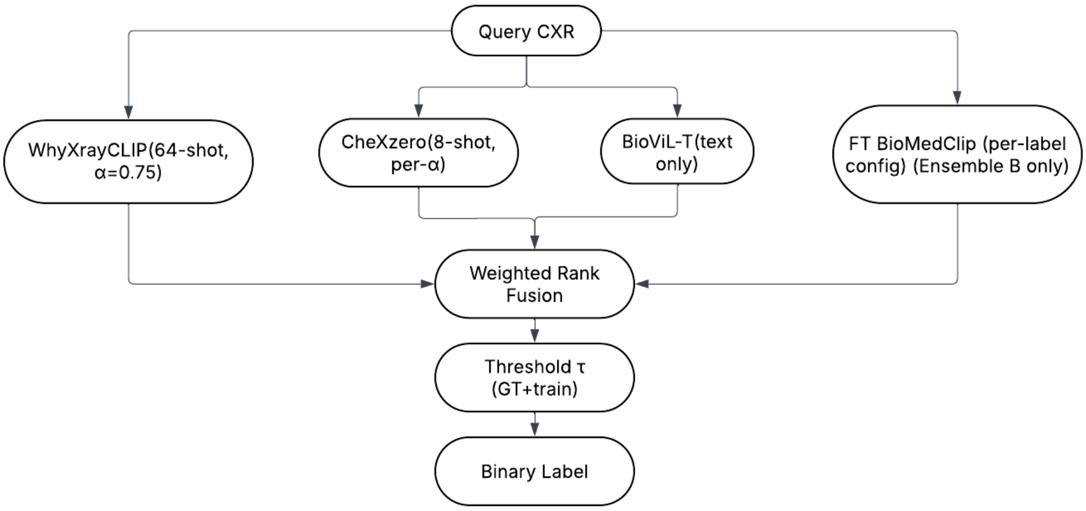

Figure 4: Ensemble inference pipeline. Ensemble A uses the top three models (WhyXray-CLIP, CheXzero, BioViL-T); Ensemble B adds FT-BioMedCLIP. Scores are converted to ranks and fused with performance-derived weights.

6.3.2 Ensemble Benchmarking Results

Table 17 reports the full Ensemble A vs. Ensemble B comparison on the GT-set. Cells shaded in blue indicate the winner for that metric–label pair.

Table 17: Ensemble A vs. Ensemble B benchmark on GT-set. Blue = winner for that metric.

Label	Metric	Ens A	Ens B	Δ(B−A)	Winner
Edema	AUC	0.859	0.897	+0.038	Ens B
	AUPRC	0.876	0.917	+0.041	Ens B
	BAC	0.721	0.760	+0.039	Ens B
	F1	0.720	0.769	+0.049	Ens B
	Sens	0.692	0.769	+0.077	Ens B
	Spec	0.750	0.750	0.000	Tie
Atelectasis	AUC	0.811	0.826	+0.015	Ens B
	AUPRC	0.838	0.846	+0.008	Ens B
	BAC	0.691	0.662	−0.029	Ens A
	F1	0.755	0.745	−0.010	Ens A
	Sens	0.841	0.864	+0.023	Ens B
	Spec	0.541	0.459	−0.082	Ens A
Pleural Effusion	AUC	0.941	0.905	−0.036	Ens A
	AUPRC	0.912	0.850	−0.062	Ens A
	BAC	0.830	0.830	0.000	Tie
	F1	0.769	0.769	0.000	Tie
	Sens	0.909	0.909	0.000	Tie
	Spec	0.750	0.750	0.000	Tie
Enlarged Cardiomediastinum	AUC	0.865	0.844	−0.021	Ens A
	AUPRC	0.869	0.834	−0.035	Ens A
	BAC	0.794	0.824	+0.030	Ens B
	F1	0.811	0.833	+0.022	Ens B
	Sens	0.882	0.882	0.000	Tie
	Spec	0.706	0.765	+0.059	Ens B
Consolidation	AUC	0.983	0.978	−0.005	Ens A
	AUPRC	0.833	0.722	−0.111	Ens A
	BAC	0.733	0.725	−0.008	Ens A
	F1	0.158	0.154	−0.004	Tie
	Sens	1.000	1.000	0.000	Tie
	Spec	0.467	0.450	−0.017	Ens A
Macro Avg.	AUC	0.892	0.890	−0.002	Tie
	AUPRC	0.866	0.834	−0.032	Ens A
	BAC	0.754	0.760	+0.006	Ens B
	F1	0.643	0.654	+0.011	Ens B
	Sens	0.865	0.885	+0.020	Ens B
	Spec	0.643	0.635	−0.008	Ens A

Overall, both ensembles achieve competitive performance. Ensemble B shows improvements in Edema across all metrics and also improves F1 and BAC in Enlarged Cardiomediastinum. Ensemble A retains advantages in Pleural Effusion (AUC, AUPRC), Consolidation (AUPRC, BAC), and Specificity. Both ensembles achieve macro-average AUC of ≈ 0.89.

6.4 Strategies 3 & 4: Selective Per-Label Ensembles

Strategies 3 and 4 employ a selective per-label ensemble, where only the strongest-performing models are retained for each label to reduce negative contributions while preserving complementary strengths.

Table 18: Per-label model inclusion for selective ensembles (Strategies 3–4). ✓ = included; × = excluded.

Label	**WhyXrayCLIP**	**CheXzero**	**BioViL-T**	**FT-BioMedCLIP**
Edema	✓	✓	×	✓
Atelectasis	✓	✓	✓	×
Pleural Effusion	✓	✓	✓	×
Enl. Cardiomediastinum	✓	×	×	✓
Consolidation	✓	✓	✓	✓

Both strategies use the same selective model inclusion (Table 18) and the same performance-weighted rank fusion mechanism described earlier. The difference lies in how thresholds are calibrated:

- **Strategy 3 (GT-set-only calibration):** Thresholds are optimized directly on GT-set uncertain samples. For imbalanced labels (positive rate < 0.15), the highest threshold that retains all true positives is selected (zero false-negative policy, maximizing sensitivity). For more balanced labels, thresholds are chosen to maximize balanced accuracy.
- **Strategy 4 (train-set-only calibration):** Thresholds are calibrated using 600 balanced certain samples (300 per class) from the training set. A half-split cross-validation procedure is applied: one half scans thresholds to maximize balanced accuracy, while the other half validates candidate thresholds (best, midpoint, and median-positive), selecting the best-performing threshold.

Both strategies are evaluated on the full GT-set uncertain cohort ($n = 234$).

Table 19: Selective Ensemble benchmark on GT-set ($n = 234$ uncertain samples).

Label	**AUC**	**AUPRC**	**BAC**	**F1**	**Sens**	**Spec**	ΔAUC	n
Edema	0.936	0.948	0.840	0.846	0.846	0.833	−0.019	25
Atelectasis	0.826	0.846	0.662	0.745	0.864	0.459	+0.053	81
Pleural Effusion	0.950	0.930	0.805	0.741	0.909	0.700	+0.009	31
Enl. Cardiomediastinum	0.848	0.851	0.794	0.811	0.882	0.706	−0.014	34
Consolidation	0.978	0.722	0.725	0.154	1.000	0.450	+0.006	63
Macro Avg.	**0.908**	**0.859**	**0.765**	**0.659**	**0.900**	**0.630**	+0.027	**234**

Table 19 reports the resulting performance of the selective ensemble. The approach achieves a macro-AUC of 0.908 (+0.027 over the best individual model), demonstrating improved robustness. The selective inclusion also stabilizes performance across labels (e.g., excluding BioViL-T for Edema and FT-BioMedCLIP for Atelectasis), reducing per-label degradation compared to full ensembles.

7 Task 3: Relabeling the CheXpert Training Set

Following GT-set benchmarking, the best-performing ensemble configuration is used to relabel the uncertain (-1) entries for the five target labels across the entire CheXpert training set. For each uncertain sample, the ensemble produces a fused score that is compared against the calibrated threshold to assign a binary label (0 or 1). The resulting dataframe (`train_relabeled_v2.csv`) replaces all -1 entries for the five target labels with VLM-resolved binary labels, while preserving all existing 0 and 1 labels unchanged. This relabeled training set is subsequently used for downstream classifier training.

8 Task 4: Downstream Classification

8.1 Experimental Setup

To measure the downstream impact of VLM-resolved labels, we trained multi-label chest pathology classifiers and compared five uncertainty-handling strategies:

- **U-Ignore**: Uncertain samples removed from training.
- **U-Zeros**: All uncertain labels mapped to 0.
- **U-Ones**: All uncertain labels mapped to 1.
- **U-MultiClass**: Uncertain labels treated as a third class in a 3-way cross-entropy formulation.
- **U-VLM**: Uncertain labels replaced with VLM ensemble predictions (ours).

Data Split Strategy: The CheXpert `valid.csv` serves as the final held-out test set. The main `train.csv` is split 90/10 into an internal training set and an internal validation set. All strategy-specific relabeling is applied only to the training portion after the split.

8.2 Classification Architectures

Two standard deep learning architectures were evaluated:

- **DenseNet-121**: A densely connected convolutional network with ImageNet pre-training; the final fully-connected layer is replaced by a 5-output multi-label head.
- **EfficientNet-B4**: A compound-scaled convolutional architecture offering better parameter efficiency than DenseNet-121.

Both architectures were trained with masked Binary Cross-Entropy loss (for U-Ignore, U-Zeros, U-Ones, U-VLM) or multi-class cross-entropy (for U-MultiClass), warmup (2 epochs) + cosine learning rate decay, and early stopping based on validation AUROC.

8.3 Threshold Calibration for Downstream Classifiers

After training, decision thresholds for each classifier's output probabilities are calibrated using a **standard balanced-accuracy maximisation** on the internal validation set. The CheXpert training set certain-label samples are used for this calibration; no GT-set samples are used at this stage, as the GT-set serves exclusively as the VLM benchmarking reference. The final reported metrics are computed on the held-out `valid.csv` test set.

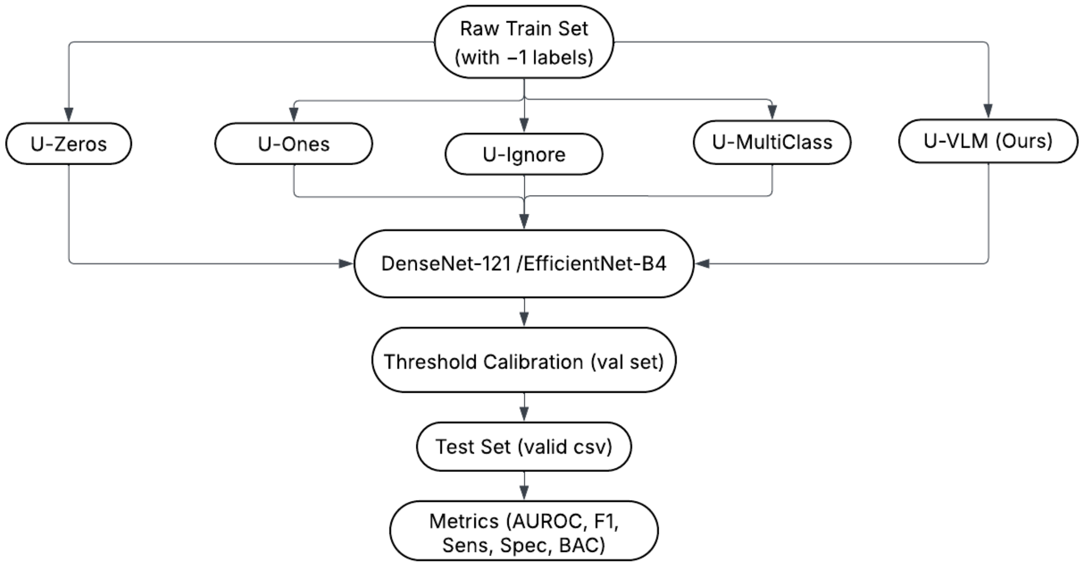

Figure 5: Downstream classification pipeline. Five uncertainty-handling strategies produce five distinct training sets. Both DenseNet-121 and EfficientNet-B4 are trained under each strategy. Decision thresholds are calibrated on the internal validation set, and final metrics are reported on the held-out CheXpert `valid.csv` test set.

8.4 Evaluation Metrics

Models are evaluated on the held-out test set using: AUROC, AUPRC, F1-Score, Sensitivity (Recall), Specificity, and Balanced Accuracy — reported per label and averaged across the five target labels.

9 Classification Results: VLM Strategy Comparison and Baseline Analysis

In this section we compare the downstream classification performance of the four VLM ensemble strategies (Strategy 1 through Strategy 4) against the four standard uncertainty-handling baselines (U-Ignore, U-Zeros, U-Ones, U-MultiClass). Results are reported separately for DenseNet-121 and EfficientNet-B4, evaluated on the held-out `valid.csv` test set across the five target labels.

9.1 DenseNet-121: VLM Strategy Comparison

Tables 20 and 21 compare AUROC and AUPRC across the three DenseNet VLM strategies. For Strategy 2, we report Ensemble B results (the higher-performing four-model variant; Ensemble A achieved a macro-average AUROC of 0.8748 vs. 0.8724 for Ensemble B on the five target labels).

Table 20: DenseNet-121 AUROC for 5 target labels across VLM strategies. **Bold** = highest per row.

Label	S1	S2-EnsA	S2-EnsB	S3	S4
Edema	0.9149	0.9268	0.9250	**0.9434**	0.9285
Atelectasis	0.8593	0.8577	**0.8705**	0.8606	0.8459
Pleural Effusion	0.9266	**0.9267**	0.9247	0.9201	0.9188
Enl. Cardiomediastinum	**0.7588**	0.7099	0.7109	0.6980	0.6789
Consolidation	0.9218	**0.9528**	0.9310	0.9170	0.9432
Mean (5 labels)	0.8763	0.8748	0.8724	**0.8878**	0.8821

Table 21: DenseNet-121 AUPRC for 5 target labels across VLM strategies. **Bold** = highest per row.

Label	S1	S2-EnsA	S2-EnsB	S3	S4
Edema	0.7549	0.7570	0.7592	**0.8015**	0.7609
Atelectasis	0.7512	0.7494	0.7442	**0.7691**	0.7121
Pleural Effusion	**0.8376**	0.8331	0.8282	0.8299	0.8179
Enl. Cardiomediastinum	**0.7233**	0.6678	0.6503	0.6638	0.6350
Consolidation	0.5976	**0.6967**	0.6497	0.6220	0.5995
Mean (5 labels)	0.7329	**0.7408**	0.7263	0.7373	0.7051

Best DenseNet Strategy. By macro-average AUROC, **Strategy 3** achieves the highest score (0.8878), outperforming Strategy 1 (0.8763), Strategy 2 Ensemble A (0.8748), and Strategy 4 (0.8821). By macro-average AUPRC, Strategy 2 Ensemble A leads narrowly (0.7408) over Strategy 3 (0.7373), a gap of only 0.0035. Since AUROC is the primary evaluation metric in the CheXpert literature and the two metrics are in near-agreement, we adopt **Strategy 3** as the primary DenseNet VLM method for comparison against baselines. Strategy 3's selective per-label ensemble with GT-set-calibrated thresholds achieves consistent gains across Edema, Atelectasis, Pleural Effusion, and Consolidation while the marginal AUPRC deficit relative to Ensemble A is negligible in practice.

9.2 DenseNet-121: VLM vs. Standard Baselines

Table 22: DenseNet-121 AUROC: Strategy 3 U-VLM vs. standard baselines. Green = best per row.

Label	U-Ignore	U-Zeros	U-Ones	U-MultiClass	U-VLM (S3)
Edema	0.9268	0.9103	0.9247	0.9070	**0.9434**
Atelectasis	0.8069	0.8115	0.8187	0.7780	**0.8606**
Pleural Effusion	0.9212	0.9255	0.9207	**0.9352**	0.9201
Enl. Cardiomediastinum	0.5210	0.5981	0.5213	0.6309	**0.6980**
Consolidation	0.9085	0.9007	0.8634	**0.9136**	0.9170
Mean (5 labels)	0.8169	0.8292	0.8098	0.8329	**0.8878**

Table 23: DenseNet-121 AUPRC: Strategy 3 U-VLM vs. standard baselines. Green = best per row.

Label	U-Ignore	U-Zeros	U-Ones	U-MultiClass	U-VLM (S3)
Edema	0.7568	0.7206	0.7696	0.6955	**0.8015**
Atelectasis	0.6567	0.6773	0.6761	0.5988	**0.7691**
Pleural Effusion	0.8343	0.8370	0.8195	**0.8597**	0.8299
Enl. Cardiomediastinum	0.5140	0.5663	0.4930	0.5923	**0.6638**
Consolidation	0.5641	0.5599	0.4708	0.5774	**0.6220**
Mean (5 labels)	0.6652	0.6722	0.6458	0.6647	**0.7373**

DenseNet Analysis. The U-VLM strategy (Strategy 3) achieves the highest macro-average AUROC (0.8878) among all five strategies, surpassing the best baseline (U-MultiClass at 0.8329 AUROC) by +0.055 AUROC. It also leads on macro-average AUPRC (0.7373) over the best baseline (U-MultiClass at 0.6647) by +0.073 AUPRC. The most striking gain is on Enlarged Cardiomediastinum, where U-VLM achieves AUROC 0.698 vs. the best baseline 0.631 (U-MultiClass), a +0.067 improvement — the label with the highest clinical difficulty in this set. U-VLM also dominates on Edema (0.943 vs. best baseline 0.927) and Atelectasis (0.861 vs. 0.819). On Pleural Effusion, U-MultiClass marginally leads in AUROC (0.935 vs. 0.920) and AUPRC (0.860 vs. 0.830), and similarly for Consolidation in AUROC (0.914 vs. 0.917), suggesting these two labels are already well-served by existing heuristics. Nonetheless, U-VLM dominates on average across all five labels, confirming that the selective per-label ensemble with GT-calibrated thresholds provides consistently superior training signal.

9.3 EfficientNet-B4: VLM Strategy Comparison

Tables 24 and 25 compare all four VLM strategies for EfficientNet-B4.

Table 24: EfficientNet-B4 AUROC for 5 target labels across all VLM strategies. **Bold** = highest per row.

Label	S1	S2-EnsA	S2-EnsB	S3	S4
Edema	0.917	**0.940**	0.930	0.934	0.936
Atelectasis	**0.841**	0.834	0.841	0.843	0.837
Pleural Effusion	**0.931**	0.928	0.925	0.930	0.928
Enl. Cardiomediastinum	**0.770**	0.738	0.736	0.705	0.719
Consolidation	0.930	**0.947**	0.947	0.923	0.945
Mean (5 labels)	**0.878**	0.877	0.876	0.867	0.873

Table 25: EfficientNet-B4 AUPRC for 5 target labels across all VLM strategies. **Bold** = highest per row.

Label	S1	S2-EnsA	S2-EnsB	S3	S4
Edema	**0.780**	0.776	0.736	0.742	0.775
Atelectasis	**0.687**	0.699	0.692	0.728	0.672
Pleural Effusion	0.829	**0.830**	0.829	0.835	0.829
Enl. Cardiomediastinum	**0.720**	0.695	0.682	0.668	0.674
Consolidation	0.646	**0.653**	0.649	0.655	0.647
Mean (5 labels)	**0.732**	0.731	0.718	0.726	0.719

Best EfficientNet Strategy. By both macro-average AUROC (0.878) and macro-average AUPRC (0.732), **Strategy 1** achieves the highest scores for EfficientNet-B4 as well. The two metrics agree, confirming Strategy 1 as the overall winner. We therefore use Strategy 1 as the representative U-VLM method for EfficientNet baseline comparisons.

9.4 EfficientNet-B4: VLM vs. Standard Baselines

Table 26: EfficientNet-B4 AUROC: Strategy 1 U-VLM vs. standard baselines. Green = best per row.

Label	U-Ignore	U-Zeros	U-Ones	U-MultiClass	U-VLM (S1)
Edema	0.905	0.927	**0.933**	0.925	0.917
Atelectasis	0.806	0.780	0.810	0.781	**0.841**
Pleural Effusion	0.923	0.928	**0.935**	0.930	0.931
Enl. Cardiomediastinum	0.552	0.525	0.507	0.512	**0.770**
Consolidation	0.919	**0.927**	0.873	0.913	0.930
Mean (5 labels)	0.821	0.817	0.812	0.812	**0.878**

Table 27: EfficientNet-B4 AUPRC: Strategy 1 U-VLM vs. standard baselines. Green = best per row.

Label	U-Ignore	U-Zeros	U-Ones	U-MultiClass	U-VLM (S1)
Edema	0.721	0.737	**0.780**	0.718	0.756
Atelectasis	0.633	0.582	0.623	0.607	**0.687**
Pleural Effusion	0.822	**0.845**	0.840	0.836	0.829
Enl. Cardiomediastinum	0.541	0.507	0.474	0.492	**0.720**
Consolidation	0.608	**0.665**	0.441	0.614	0.646
Mean (5 labels)	0.665	0.667	0.632	0.653	**0.728**

EfficientNet Analysis. For EfficientNet-B4, the U-VLM strategy (Strategy 1) achieves the highest macro-average AUROC (0.878) and AUPRC (0.728) across all five strategies, outperforming the best baseline (U-Ignore at 0.821 AUROC, 0.665 AUPRC) by +0.057 AUROC and +0.063 AUPRC. The Enlarged Cardiomediastinum gain is again the most pronounced: U-VLM achieves AUROC 0.770 versus the best baseline 0.552 (U-Ignore), a +0.218 absolute improvement. On Edema, U-Ones achieves a slightly higher AUROC (0.933 vs. 0.917) and AUPRC (0.780 vs. 0.756), suggesting that for this label the positive-bias heuristic coincidentally aligns with the true distribution. Nonetheless, across all five labels U-VLM dominates on average, confirming that VLM-resolved labels provide consistently superior training signal compared to any single label-flipping baseline.

9.5 Cross-Architecture Summary

Table 28: Summary: best mean AUROC and AUPRC (5 target labels) for each approach across both architectures. Green = best per column among all strategies. The best-performing U-VLM strategy differs by architecture: Strategy 3 for DenseNet-121 and Strategy 1 for EfficientNet-B4.

Strategy	**DenseNet-121**		**EfficientNet-B4**	
	AUROC	AUPRC	AUROC	AUPRC
U-Ignore	0.817	0.665	0.821	0.665
U-Zeros	0.829	0.672	0.817	0.667
U-Ones	0.810	0.646	0.812	0.632
U-MultiClass	0.833	0.665	0.812	0.653
U-VLM (S1)	0.876	0.733	**0.878**	**0.732**
U-VLM (S2-EnsA)	0.875	**0.741**	0.877	0.730
U-VLM (S2-EnsB)	0.872	0.726	0.876	0.718
U-VLM (S3)	**0.888**	0.737	0.867	0.726
U-VLM (S4)	0.882	0.705	0.873	0.719

Across both architectures and both primary metrics, the U-VLM approaches consistently and substantially outperform all standard uncertainty-handling baselines. For DenseNet-121, the best U-VLM configuration is **Strategy 3** (selective per-label ensemble with GT-set-calibrated thresholds), achieving macro-average AUROC of 0.888 and AUPRC of 0.737, surpassing the best baseline (U-MultiClass at 0.833 AUROC, 0.665 AUPRC) by +0.055 AUROC and +0.072 AUPRC. For EfficientNet-B4, the best U-VLM configuration is **Strategy 1**, achieving macro-average AUROC of 0.878 and AUPRC of 0.728, surpassing the best baseline (U-Ignore at 0.821 AUROC, 0.665 AUPRC) by +0.057 AUROC and +0.065 AUPRC. The fact that the optimal strategy differs by architecture — Strategy 3's GT-calibrated selective ensemble benefits DenseNet-121 more, while Strategy 1's simpler three-model ensemble proves sufficient for EfficientNet-B4 — suggests that stronger architectures are less sensitive to label noise and benefit less from the more aggressive calibration in Strategy 3. Notably, S2-EnsA achieves the highest AUPRC for DenseNet-121 (0.741), only 0.004 above S3 (0.737), confirming that the two leading strategies are effectively tied on precision-recall performance. Across all configurations, U-VLM methods uniformly and substantially outperform every deterministic label-flipping baseline, validating the core hypothesis that VLM-resolved uncertain labels provide higher-quality training signal than any heuristic uncertainty-handling strategy.

10 Summary of Contributions

This project makes the following contributions:

1. **Systematic multi-model benchmarking with context ablation.** We benchmarked four medical VLMs (WhyXrayCLIP, CheXzero, BioMedCLIP, BioViL-T) on the task of uncertain label resolution against a multi-radiologist ground-truth consensus (GT-set), with ablation studies over three context-format strategies (plain text, radiology language, no context) and three threshold calibration methods. Few-shot prototype augmentation consistently outperformed zero-shot inference for WhyXrayCLIP and CheXzero, with WhyXrayCLIP achieving a mean AUC improvement of +0.158 (ZS 0.723 $\rightarrow$ FS 0.881) and CheXzero +0.149 (ZS 0.699 $\rightarrow$ FS 0.848) on the GT-set uncertain cohort.

2. **Domain-adaptive fine-tuning of BioMedCLIP with per-label model routing.** A three-stage curriculum fine-tuning pipeline for BioMedCLIP using LoRA adaptation (r=16) on MIMIC-CXR was developed, bridging the domain gap between general biomedical image-text pre-training and chest X-ray uncertainty resolution. Fine-tuning yielded large per-label gains (e.g., Enlarged Cardiomediastinum AUC +0.147, Atelectasis +0.101, Pleural Effusion +0.034) while regressing on others, motivating a per-label model routing strategy that selects the best variant per pathology. In few-shot mode, BioMedCLIP achieves AUC 0.955 on Edema and 0.972 on Consolidation.

3. **Four ensemble strategies with GT-validated benchmarking.** Four ensemble configurations were designed and evaluated: Strategy 1 (three-model ensemble with train-set calibration and GT-set consolidation patch), Strategy 2 (Ensemble A: three-model; Ensemble B: four-model, both with weighted GT+train calibration), Strategy 3 (selective per-label ensemble with GT-set-only calibration), and Strategy 4 (selective per-label ensemble with train-set-only calibration). All ensembles were validated on the 234-sample GT-set uncertain cohort.

4. **VLM-based relabeling of CheXpert and downstream classifier evaluation.** The best ensemble per architecture was used to relabel all uncertain (-1) entries for five target pathologies in the CheXpert training set, producing a fully resolved binary training corpus. Downstream DenseNet-121 and EfficientNet-B4 classifiers trained on VLM-resolved labels were evaluated against four standard uncertainty-handling baselines (U-Ignore, U-Zeros, U-Ones, U-MultiClass). The U-VLM strategy achieves macro-average AUROC of **0.888** (DenseNet-121, Strategy 3) and **0.878** (EfficientNet-B4, Strategy 1), surpassing the best baseline by +0.055 and +0.057 AUROC respectively, and +0.072 and +0.063 AUPRC respectively. The most clinically significant gain is on Enlarged Cardiomediastinum, where U-VLM improves AUROC by +0.067 (DenseNet) and +0.218 (EfficientNet) over the best baseline, confirming that VLM-resolved labels provide higher-quality training signal than any deterministic label-flipping heuristic across both architectures.

References

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